

System Analysis and Design

Final Project Report

Group 5

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Homo Connectus

Social Media app

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Diagram (draw.io)

<https://drive.google.com/file/d/1Ss0qUF3nJ3nxwIl80RpD7Z3B4WiyleVH/view?usp=sharing>

UI Prototype (figma)

<https://www.figma.com/proto/S7DbYMipLE9g9posm8sRiY/Homo-Connectus?node-id=61-721&t=Q4nPY6ZEvR9SbwG8-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>

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CHAPTER 1

INTRODUCTION

1.1 Background

Social media platforms have become an integral part of modern life. Inspired by Instagram, the Homo Connectus project aims to provide a platform for users to share moments, interact with friends, and manage their personal data in an engaging way. This application will focus on user interactions such as posting content, following friends, and engaging with posts via likes and comments. It addresses the increasing demand for easy-to-use and secure social platforms.

1.2 Objective

The goal of this project is to design and implement Homo Connectus, a social media platform inspired by Instagram. The application will be designed using object-oriented principles and will feature:

1. User authentication (login and registration).
2. Post creation and sharing (photos, text).
3. Interaction features (like, comment, follow).
4. User profile management.
5. A user-friendly UI design that mimics social media design principles.

1.3 Benefits

1. User engagement: Provides a platform for users to share and interact with content.
2. Skill development: Enhances team skills in system analysis, design, and software development.
3. Real-world application: Simulates a popular social media platform, preparing the team for real-world software design tasks.
4. System optimization: Allows for iterative testing and enhancement of features.

1.4 Output

1. Fully functioning Homo Connectus application prototype.
2. Flowchart, Class Diagram, Sequence Diagram, Activity Diagram, Use Case, and Scenario for project documentation.
3. UI mockups designed using Figma.

CHAPTER 2 DIAGRAM

2.1 Use Case Diagram

The use case diagram represents the functionalities of the system from the user's perspective. It identifies actors (users, admins) and the key actions they can perform, such as logging in, creating a post, or liking content.

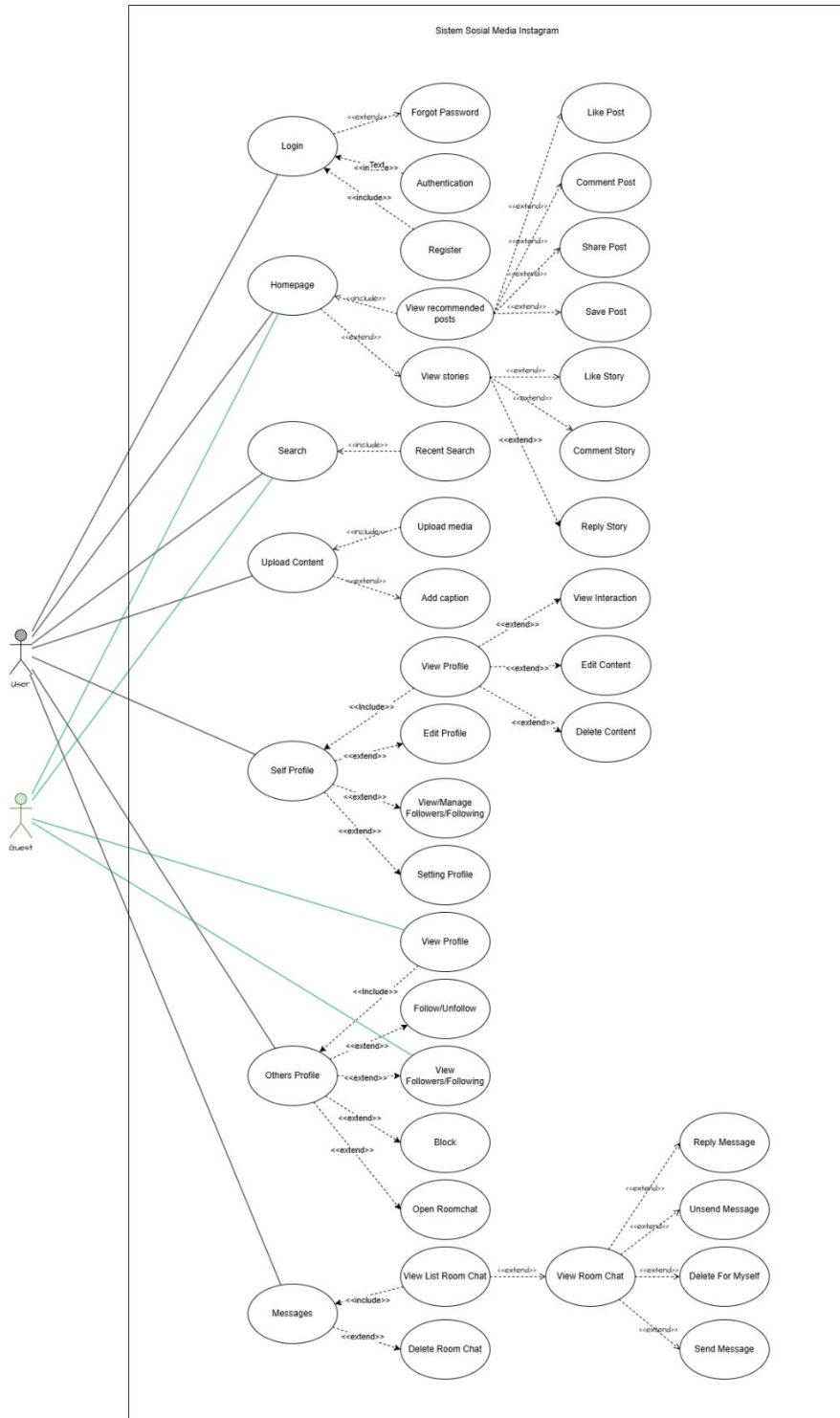


Figure 2. 1 Use Case Diagram

2.2 Use Case Scenario

The scenario diagram provides a real-world example of using the system. It walks through specific user actions and how the system responds at each step.

Login

Use Case Name	Login
System Name	Homo Connectus
Description	Allows registered users to log into the application.
Goal	To authenticate users and grant access to the app.
Actor	User
Trigger	User inputs their credentials and presses the login button.
Preconditions	User must have already registered.
Main Flow / Basic Flow	1. User opens the app. 2. Enters username and password. 3. System verifies credentials. 4. User is granted access and redirected to homepage.
Alternative Flows	Forgot Password option is selected. Register is selected.
Exception Flows	Incorrect credentials display an error. System timeout.
Postconditions	User is logged in and can access features.

Homepage

Use Case Name	Homepage
System Name	Homo Connectus
Description	Displays the feed with recommended posts and stories.
Goal	To present the user with current and recommended content.
Actor	User
Trigger	User successfully logs in.
Preconditions	User must be logged in.
Main Flow / Basic Flow	1. User is directed to homepage after login. 2. System loads recommended posts. 3. System loads current stories. 4. User interacts with content.
Alternative Flows	User refreshes homepage to get new posts.
Exception Flows	Feed fails to load due to connection issues.
Postconditions	User views current content on the platform.

Search

Use Case Name	Search
System Name	Homo Connectus
Description	Allows users to search for content, profiles, or hashtags.
Goal	To explore and discover new content or users.
Actor	User
Trigger	User taps the search icon.
Preconditions	User must be logged in.
Main Flow / Basic Flow	1. User accesses the search feature. 2. Inputs a query. 3. System displays relevant results. 4. Recent searches are saved.
Alternative Flows	User selects from recent searches instead of typing.
Exception Flows	Search service is temporarily unavailable.
Postconditions	User is shown results and can interact with them.

Upload Content

Use Case Name	Upload Content
System Name	Homo Connectus
Description	Allows users to upload media content with optional captions.
Goal	To enable users to share posts with followers.
Actor	User
Trigger	User presses the upload button.
Preconditions	User must be logged in and provide content.
Main Flow / Basic Flow	1. User selects 'Upload Content'. 2. Chooses media from gallery or camera. 3. Optionally adds a caption. 4. Presses 'Post'. 5. System shares the post to followers.
Alternative Flows	User can save the draft or cancel upload.
Exception Flows	Upload fails due to file type/size or network error.
Postconditions	Content is published to the user's profile.

Self Profile

Use Case Name	Self Profile
System Name	Homo Connectus
Description	Allows a user to view and manage their own profile.
Goal	To let users update and control their account details.
Actor	User
Trigger	User navigates to their profile.
Preconditions	User must be logged in.
Main Flow / Basic Flow	1. User clicks their profile icon. 2. Views posts, followers, and info. 3. Edits bio, profile picture, or settings. 4. Manages followers/following.
Alternative Flows	User may view interactions or delete content.
Exception Flows	Profile fails to load due to server error.
Postconditions	Profile is updated and visible to others.

Others Profile

Use Case Name	Others Profile
System Name	Homo Connectus
Description	Allows users to view profiles of other users.
Goal	To explore and interact with other users' content.
Actor	User
Trigger	User selects another user's profile.
Preconditions	User must be logged in.
Main Flow / Basic Flow	1. User taps on a post or search result. 2. Enters another user's profile. 3. Views posts and basic information. 4. Follows, unfollows, or blocks the user.
Alternative Flows	User may send a message from the profile.
Exception Flows	Blocked users' profiles are limited or hidden.
Postconditions	User has interacted or followed the target profile.

Messages

Use Case Name	Messages
System Name	Homo Connectus
Description	Enables private messaging between users.
Goal	To provide real-time communication between users.
Actor	User
Trigger	User clicks the messages icon.
Preconditions	User must be logged in.
Main Flow / Basic Flow	1. User opens messaging feature. 2. Selects a conversation or starts new one. 3. Sends, views, or replies to messages. 4. Views chat rooms and options to delete or unsend.
Alternative Flows	User deletes or unsends messages. User blocks sender.
Exception Flows	Messages fail to send due to network problems.
Postconditions	Messages are exchanged or managed accordingly.

2.3 Activity Diagram

The activity diagram represents the workflow or the sequence of activities in a specific use case. It is especially useful for visualizing the flow of user interactions in a specific task.

1. Activity Diagram Login

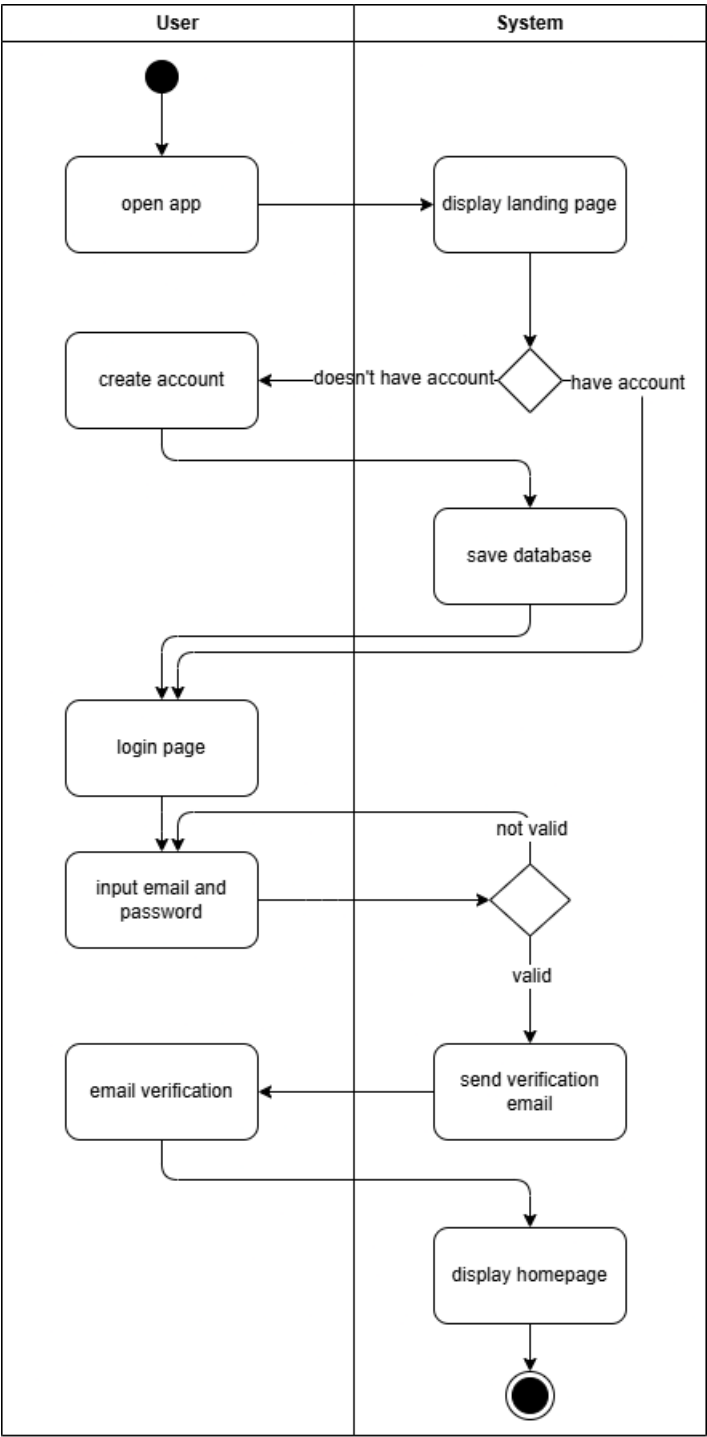


Figure 2. 2 Activity Diagram Login

2. Activity Diagram Homepage

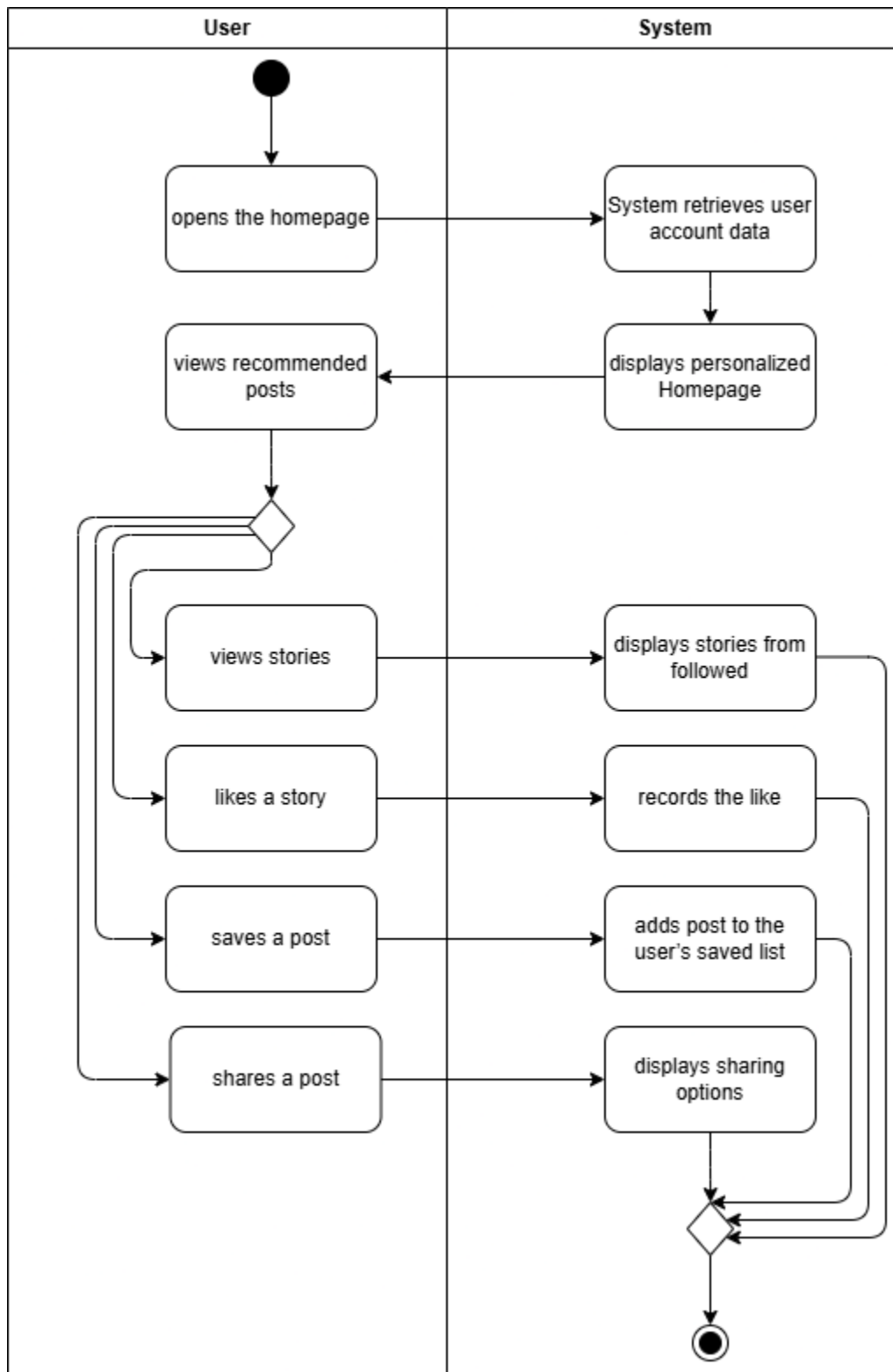


Figure 2. 3 Activity Diagram Homepage

3. Activity Diagram Search

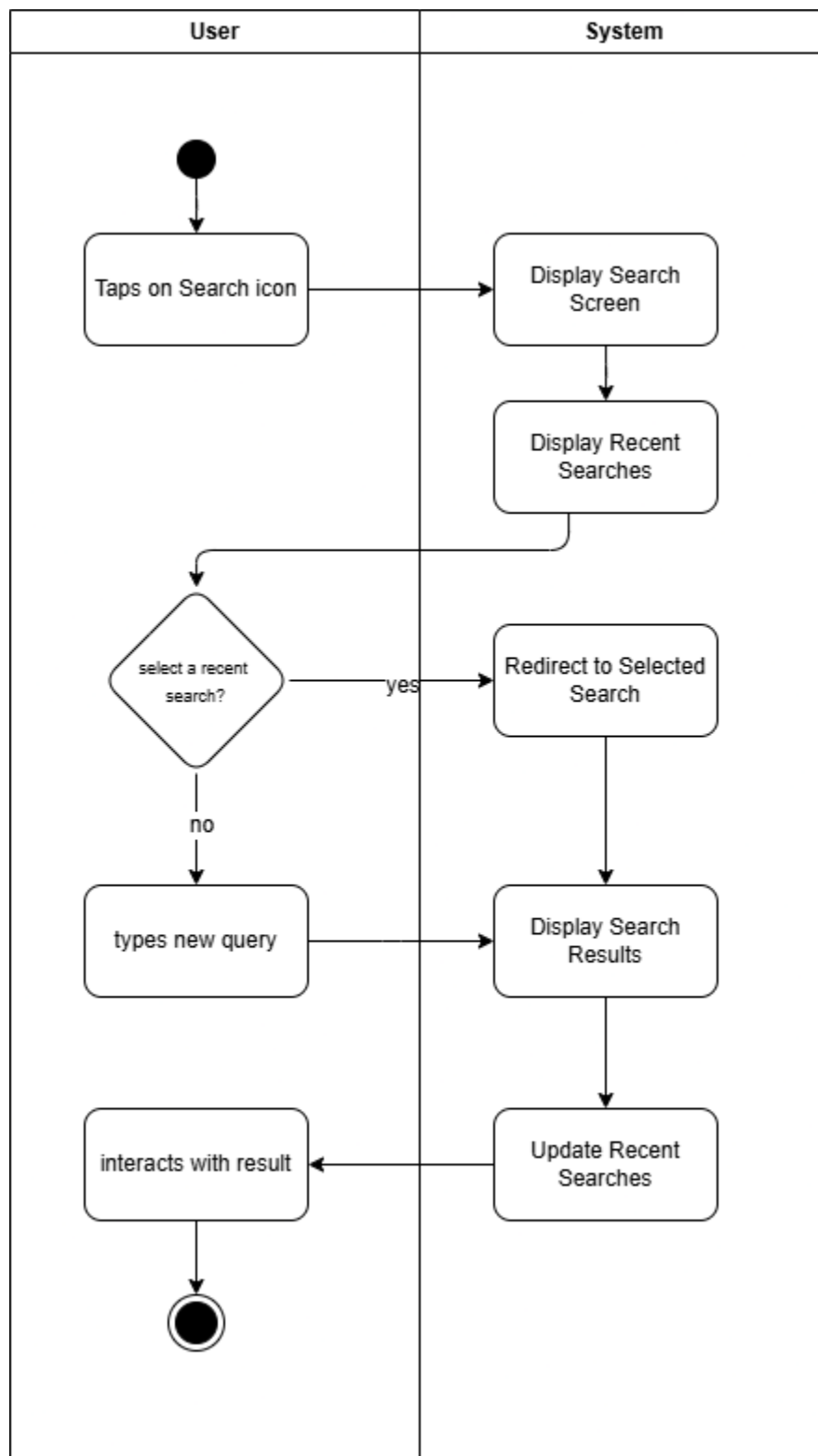


Figure 2. 4 Activity Diagram Search

4. Activity Diagram Chat

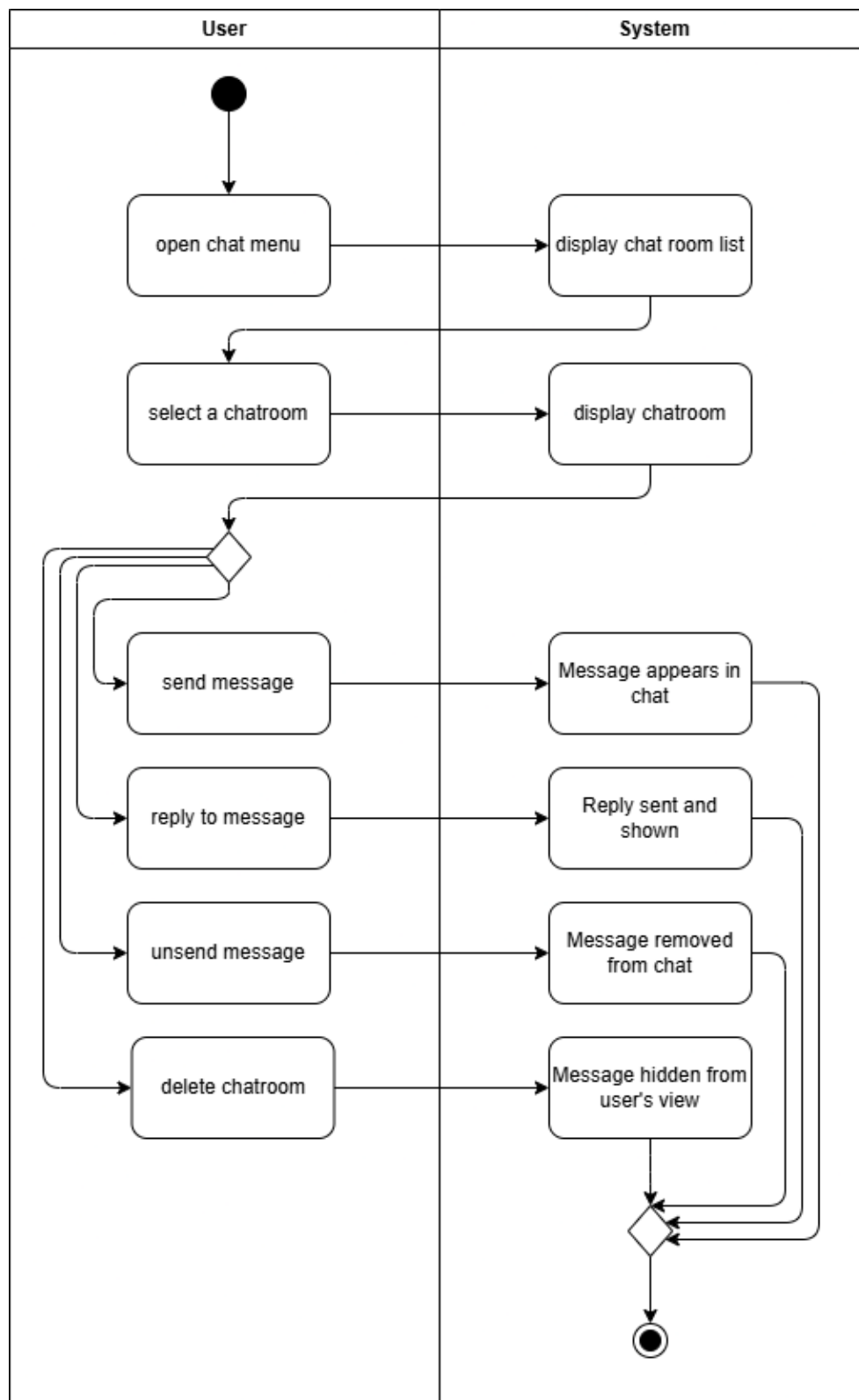


Figure 2. 5 Activity Diagram Message

5. Activity Diagram Upload

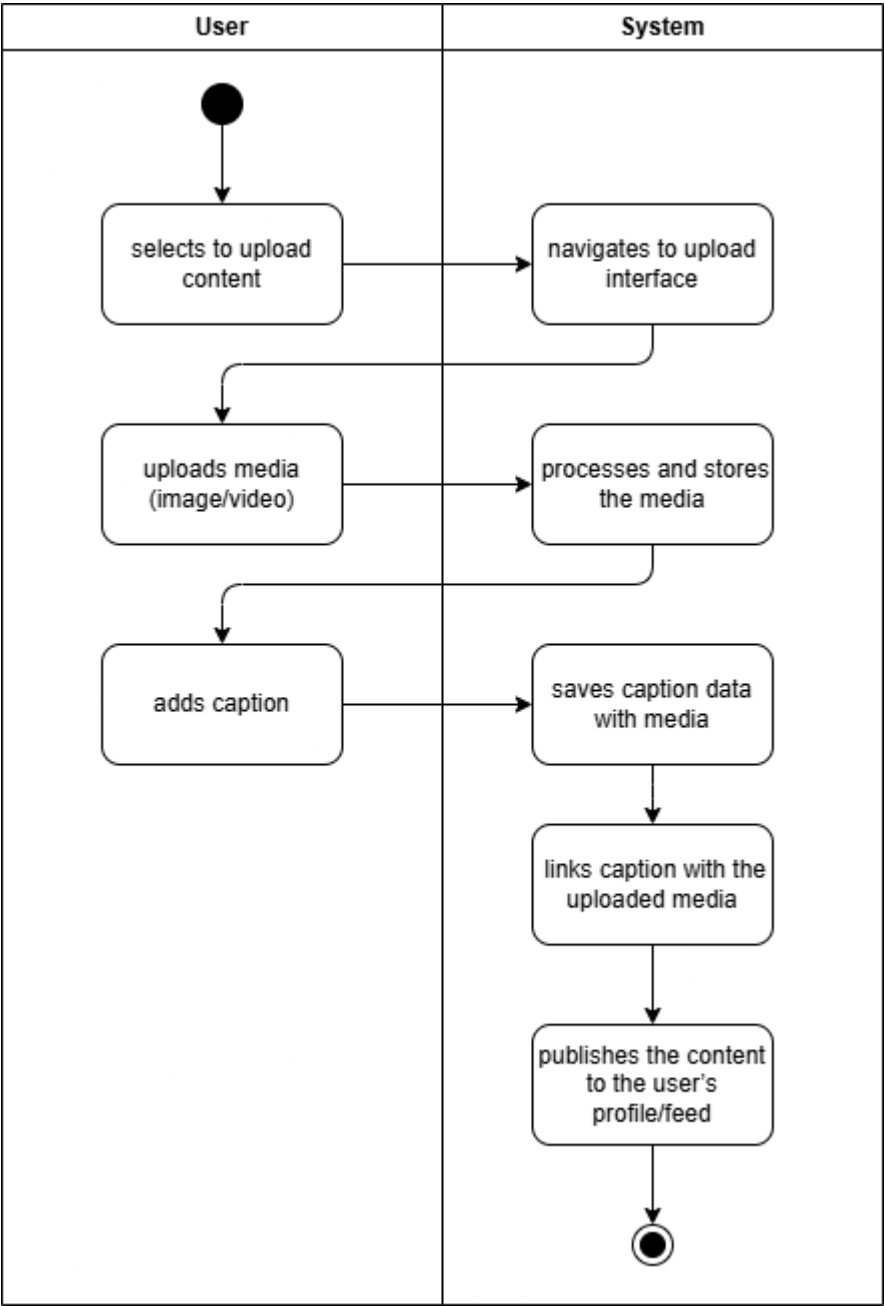


Figure 2. 6 Activity Diagram Upload

6. Activity Diagram Other's Profile

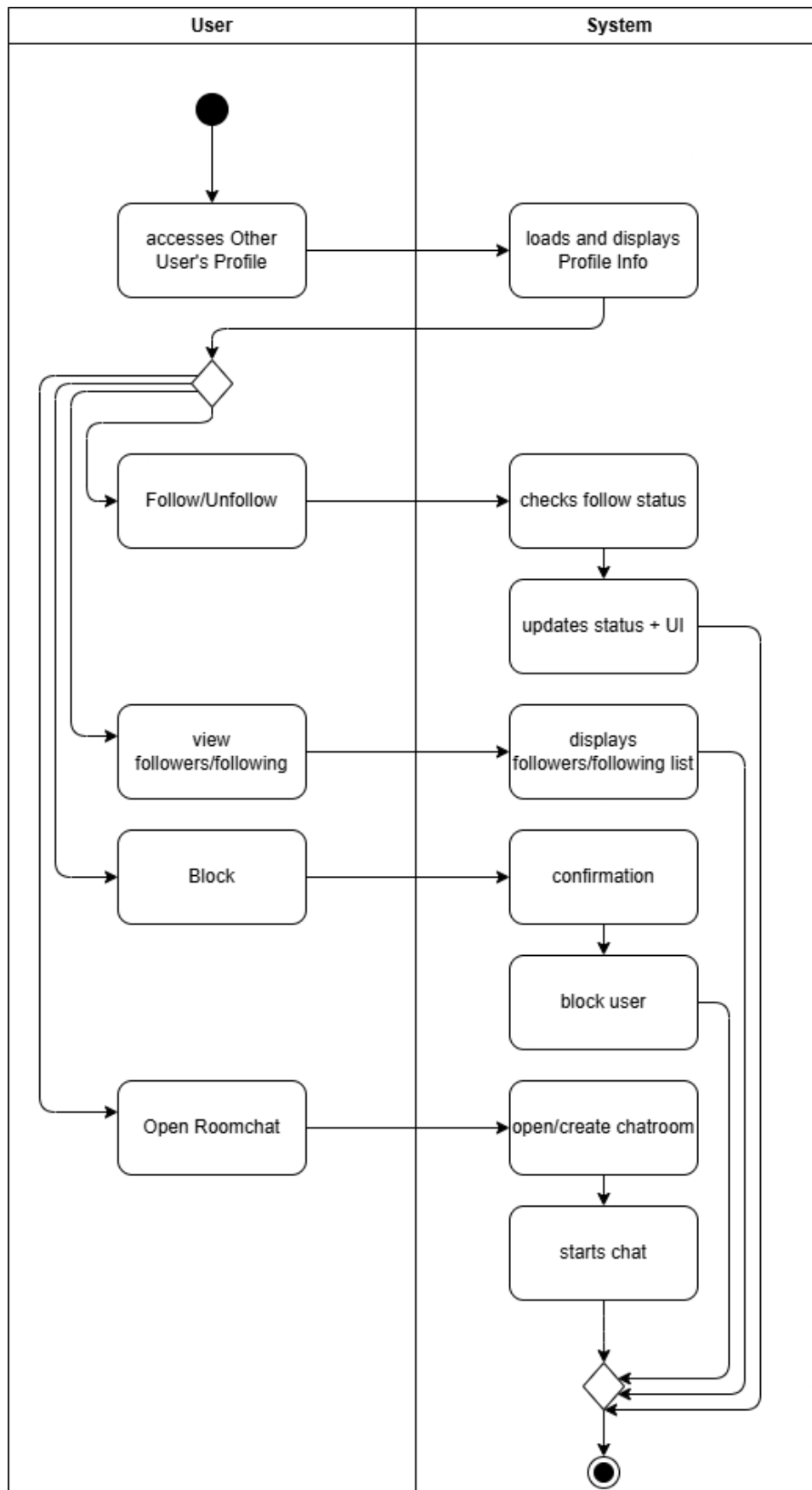


Figure 2. 7 Activity Diagram Other's Profile

7. Activity Diagram Self's Profile

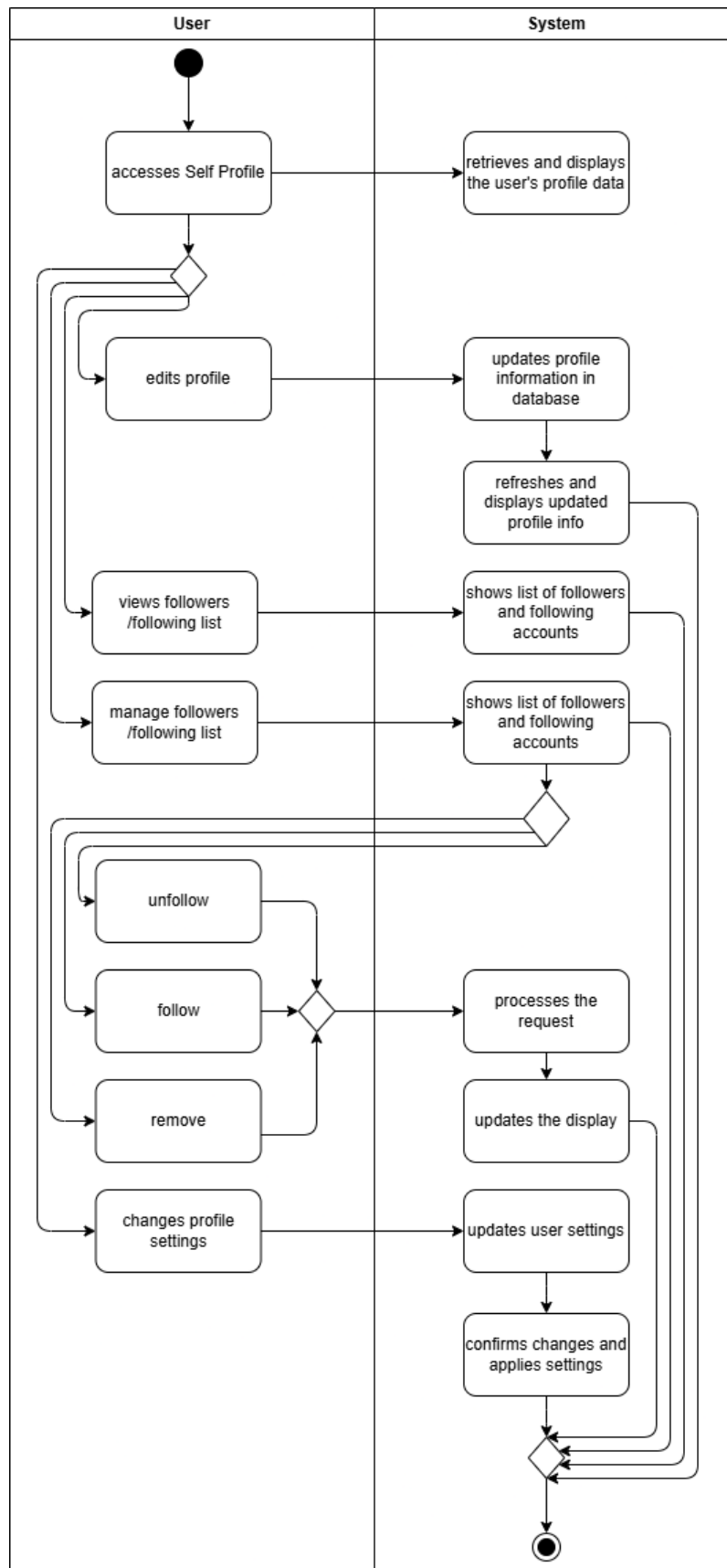


Figure 2. 8 Activity Diagram Self's Profile

2.4 Flowchart

The flowchart outlines the step-by-step process for key actions in the application, such as logging in, creating a post, or interacting with content. It helps visualize the sequence of events, decisions, and actions that a user takes to achieve their goals within the app.

1. Flowchart Login

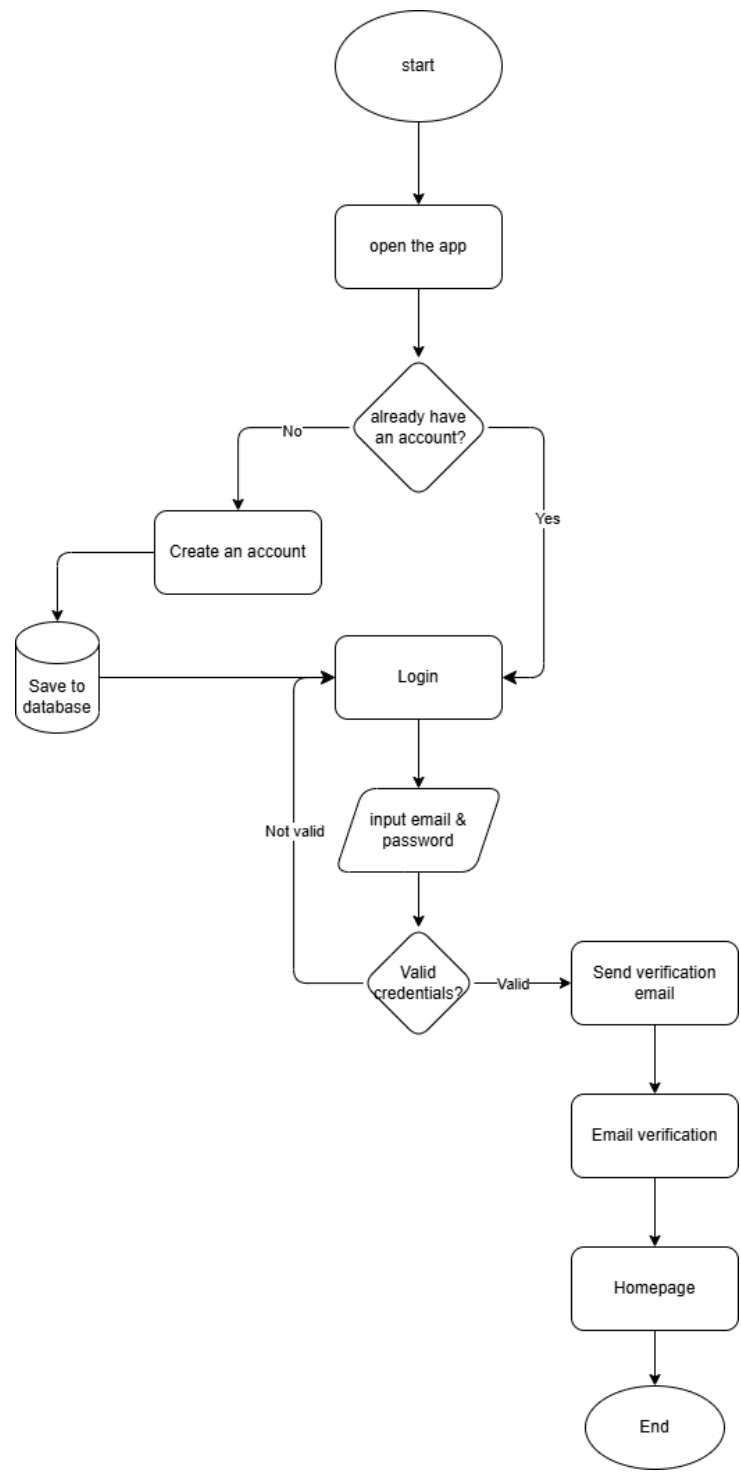


Figure 2. 9 Flowchart Login

2. Flowchart Homepage

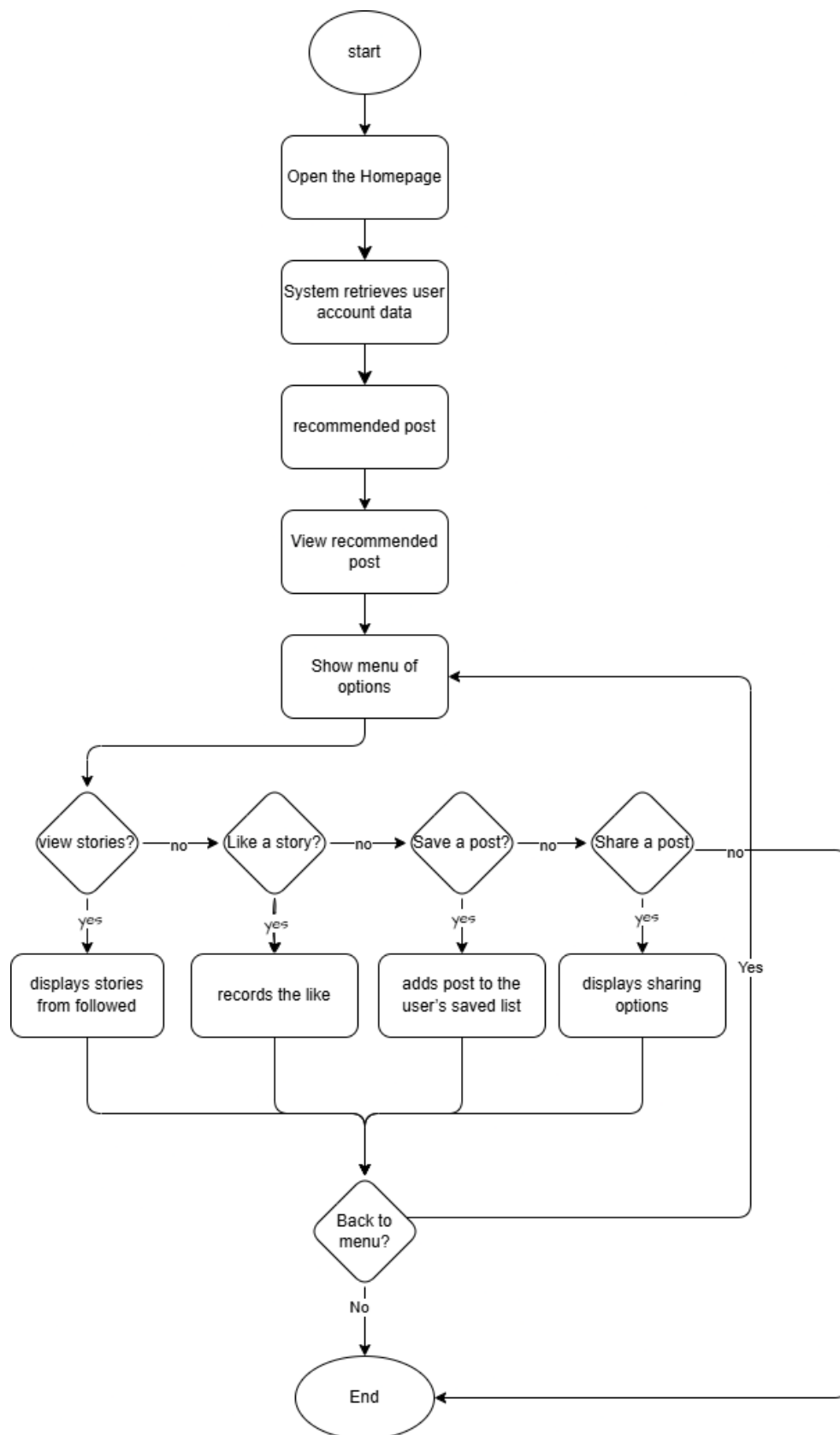


Figure 2. 10 Flowchart Homepage

3. Flowchart Upload

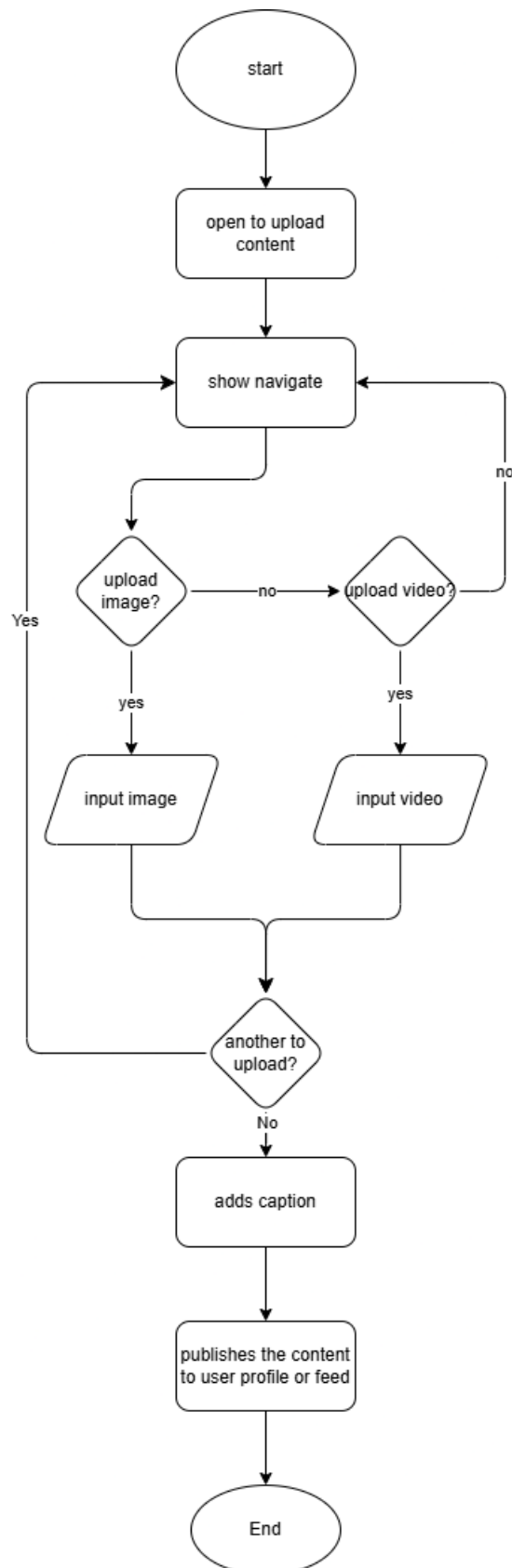


Figure 2. 11 Flowchart Upload

4. Flowchart Search

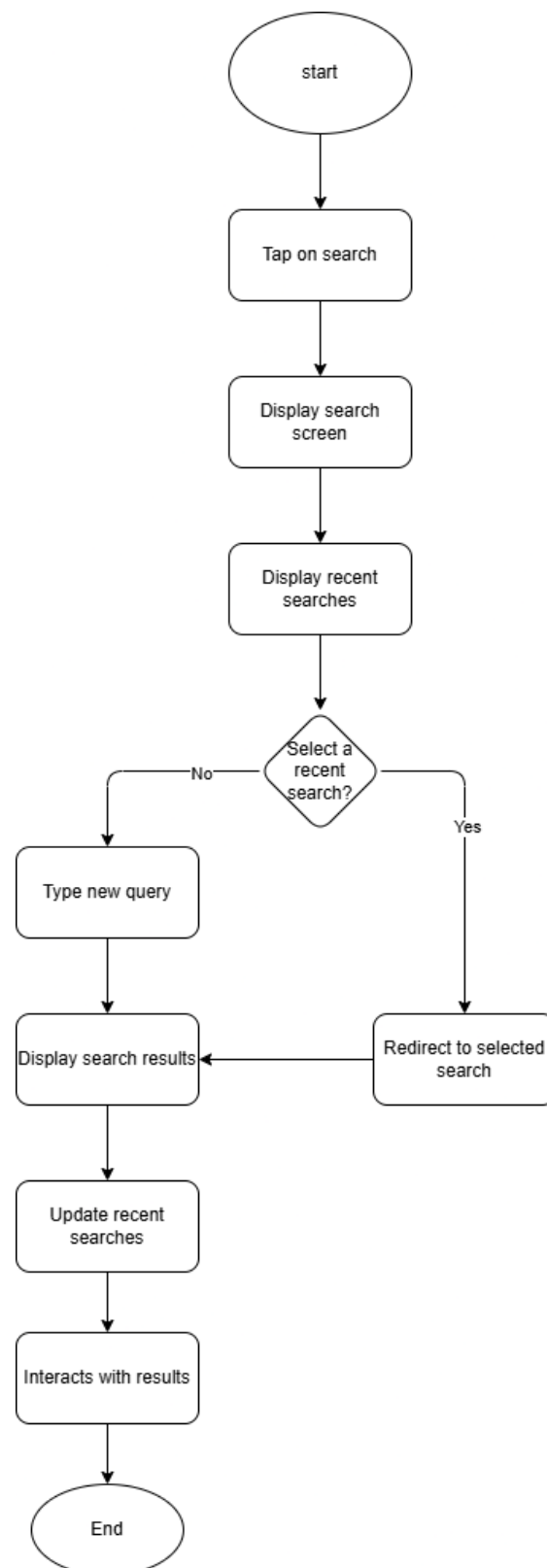


Figure 2. 12 Flowchart Search

5. Flowchart Message

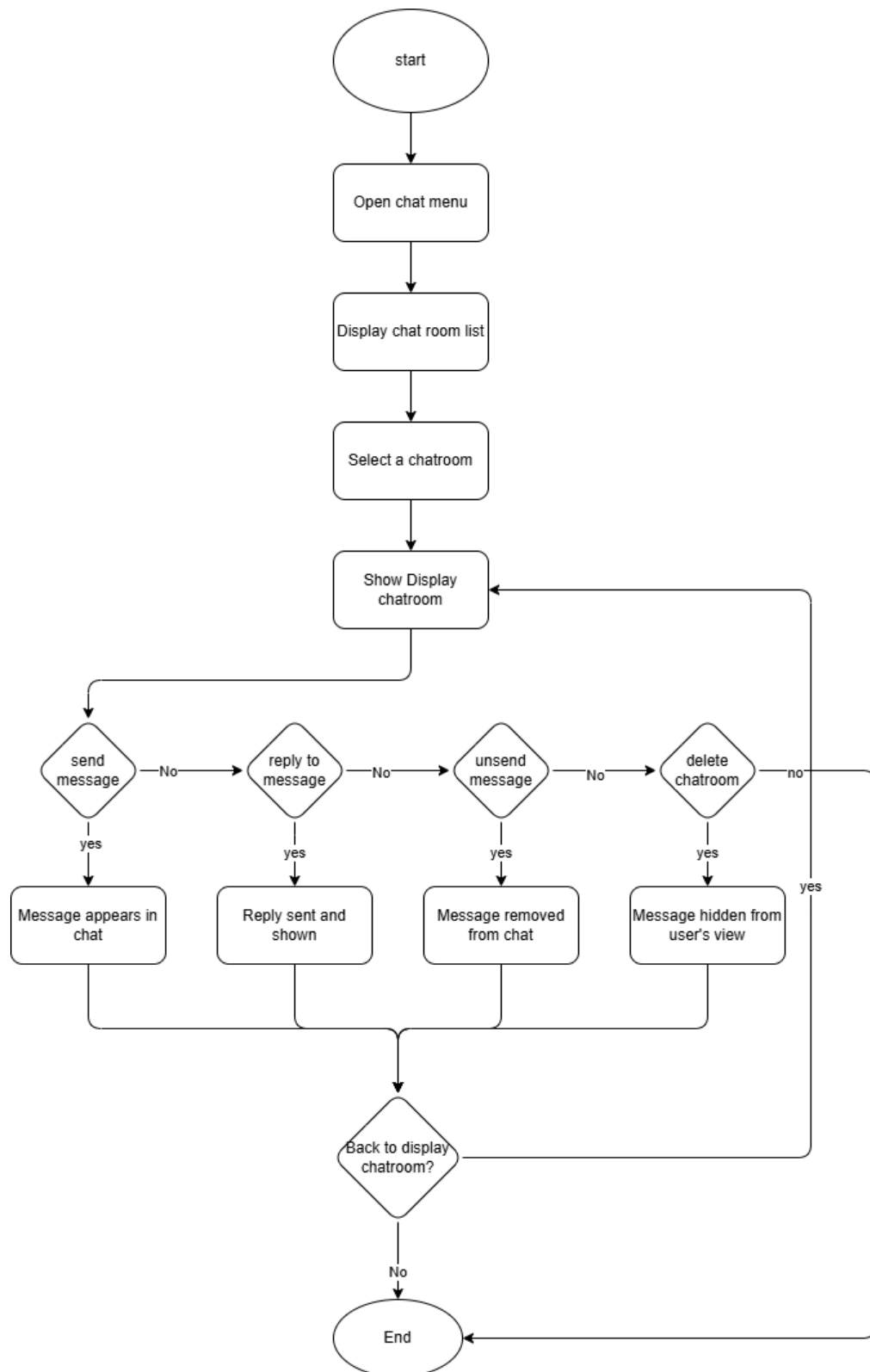


Figure 2. 13 Flowchart Message

6. Flowchart Self's Profile

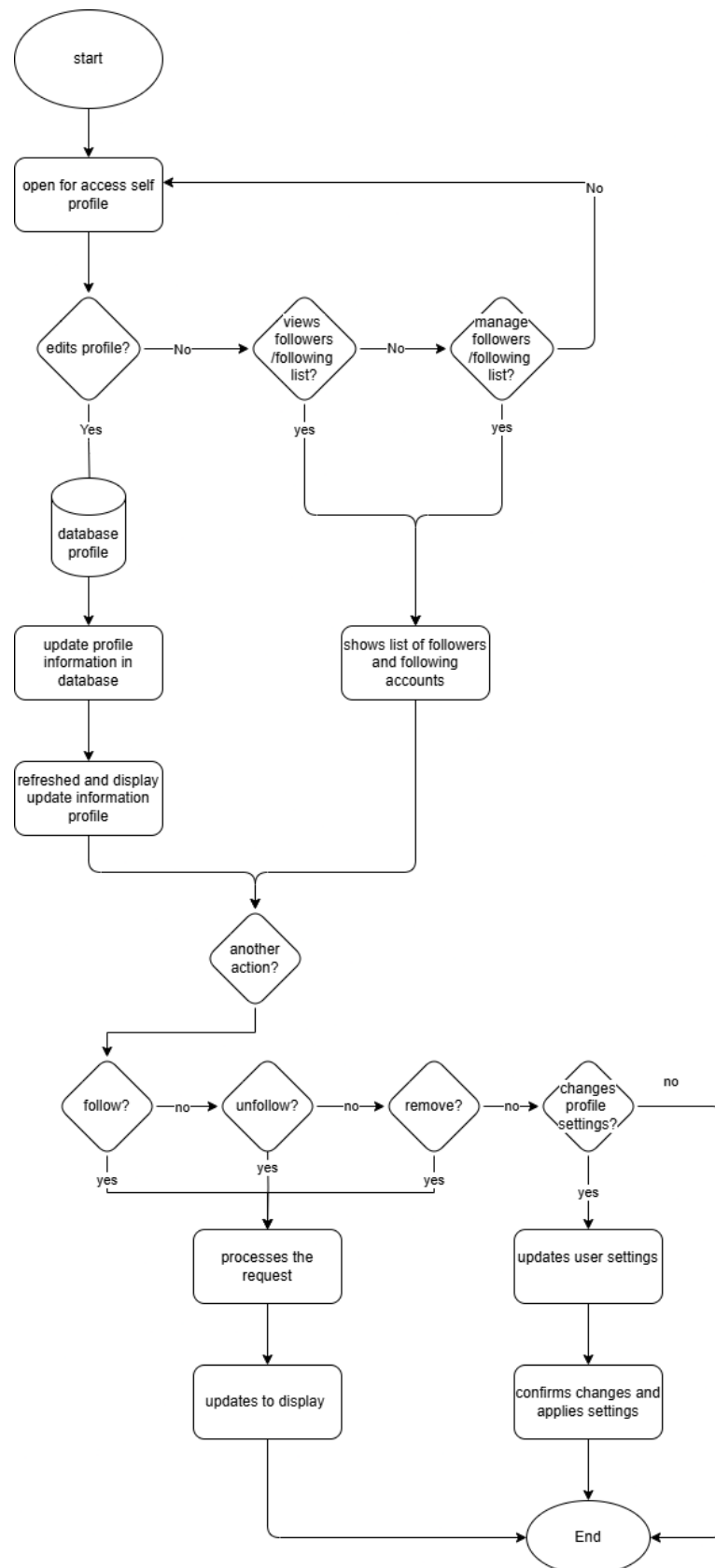


Figure 2. 14 Flowchart Self's Profile

7. Flowchart Other's Profile

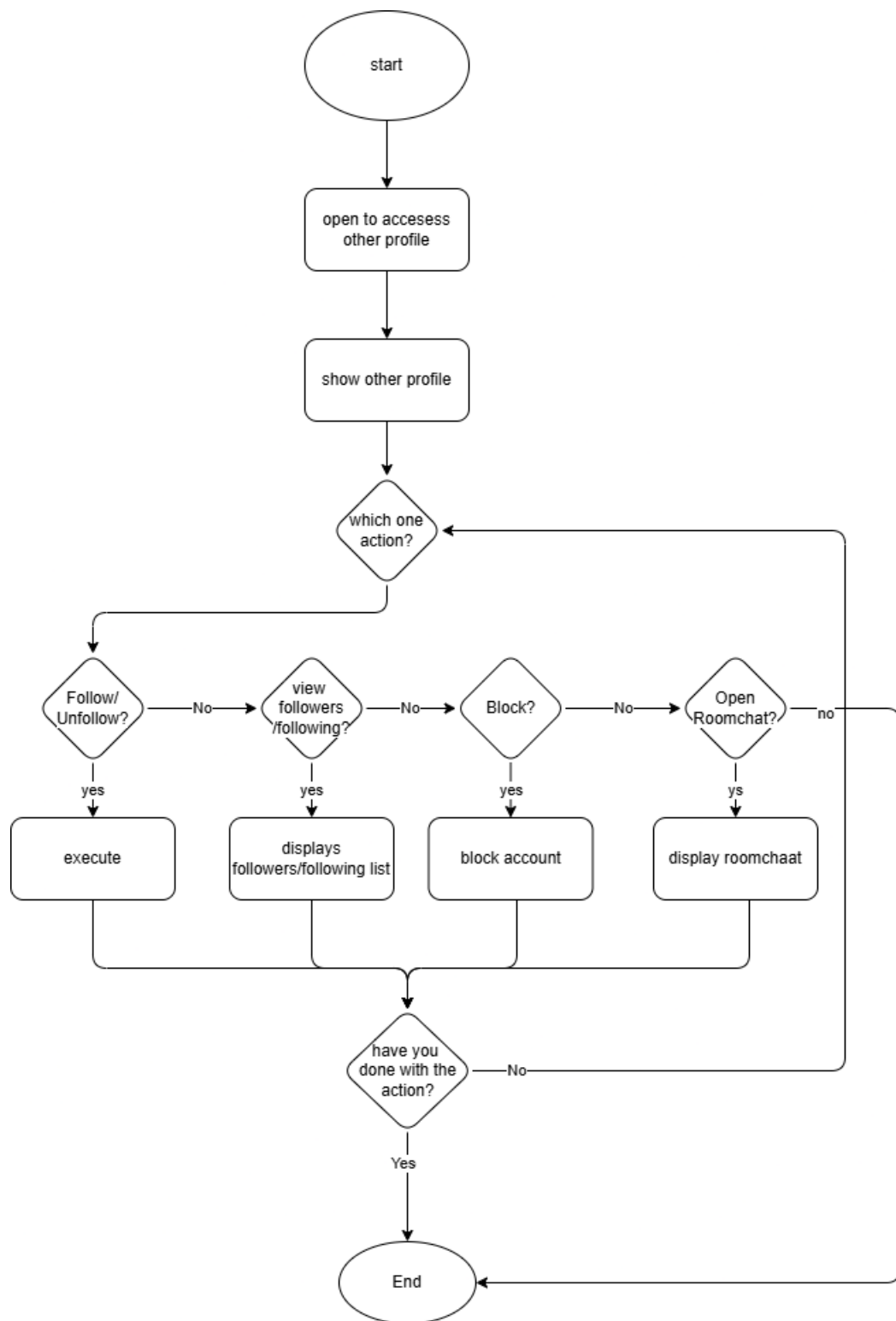


Figure 2. 15 Flowchart Other's Profile

2.5 Sequence Diagram

The sequence diagram shows the interaction between objects over time, focusing on the sequence of messages exchanged between objects to complete a specific task.

1. Sequence Login

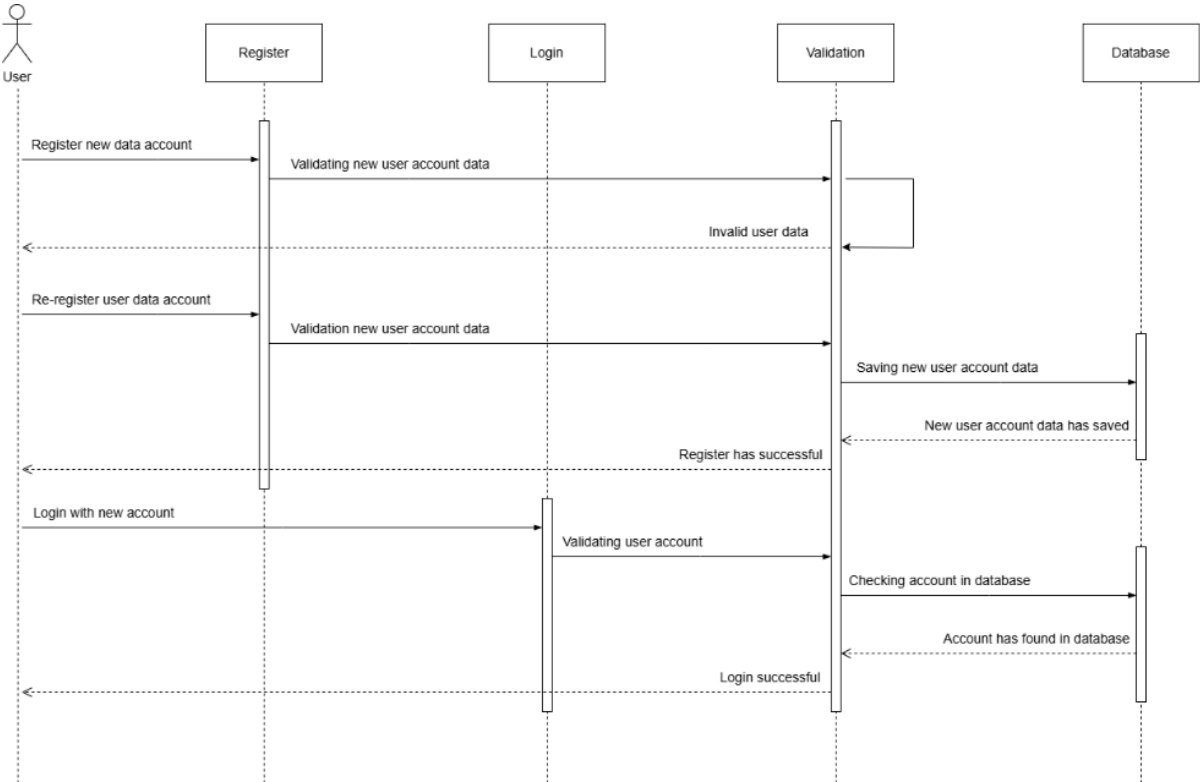


Figure 2. 16 Sequence Login

2. Sequence Homepage

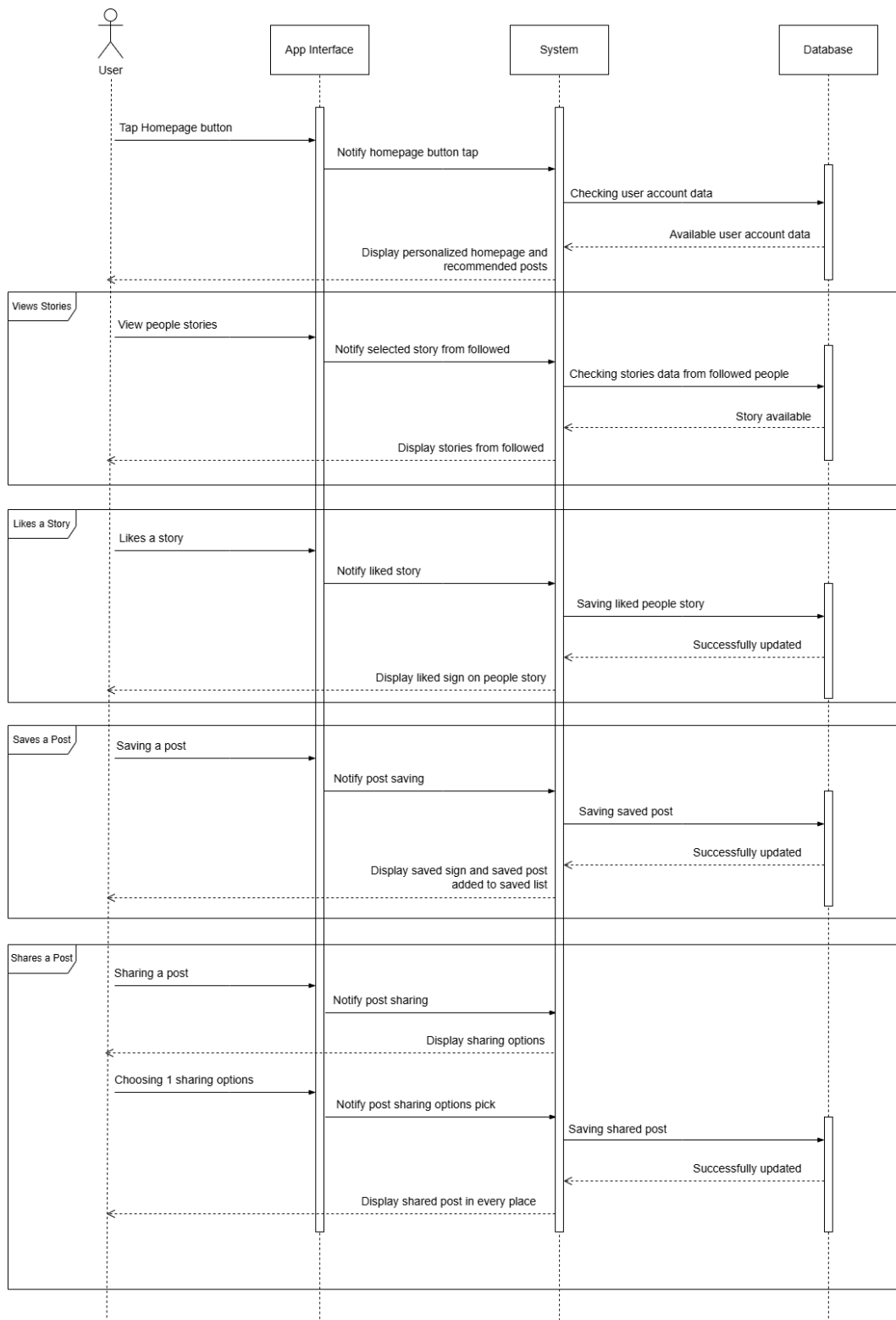


Figure 2. 17 Sequence Homepage

3. Sequence Upload

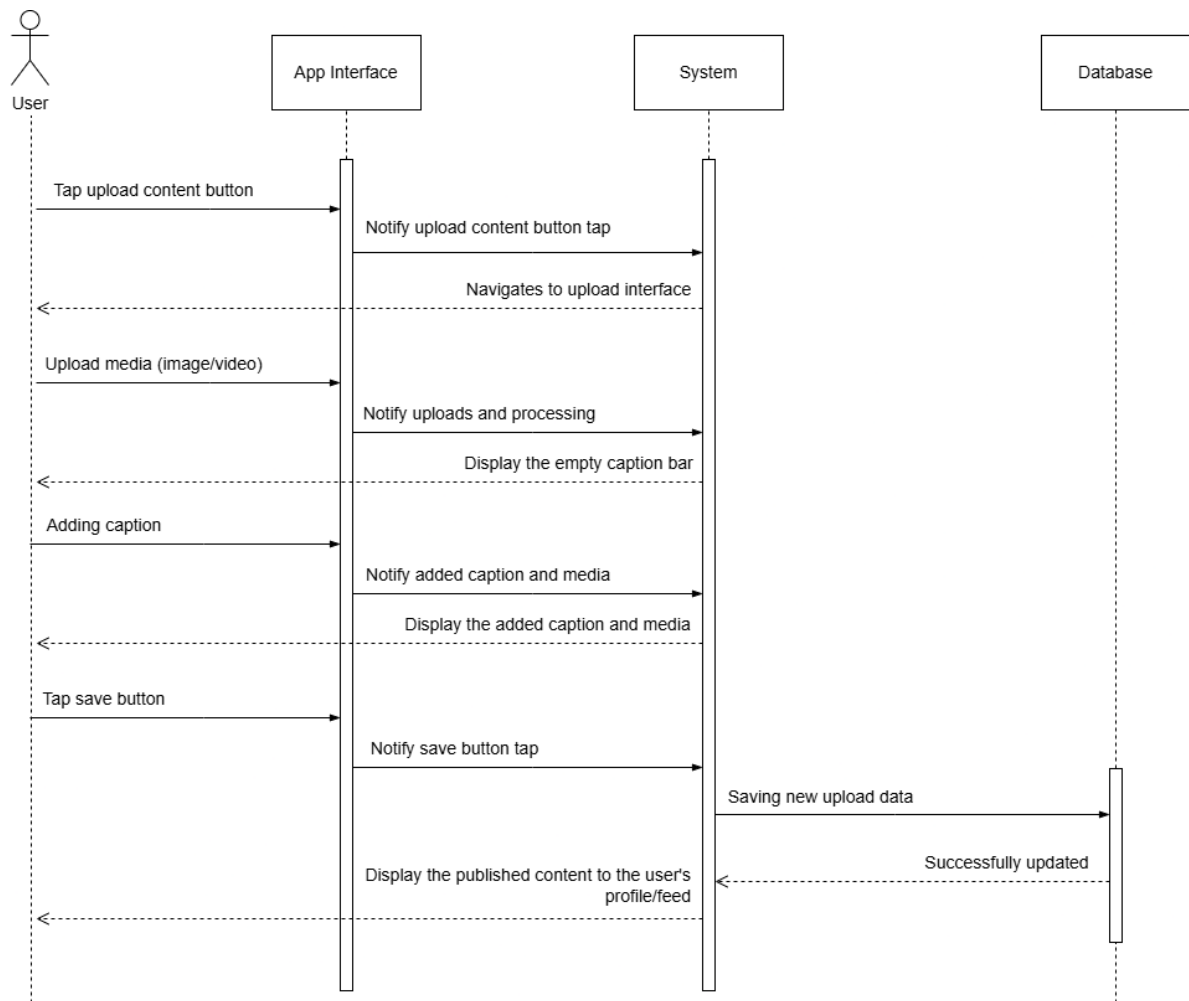


Figure 2. 18 Sequence Upload

4. Sequence Search

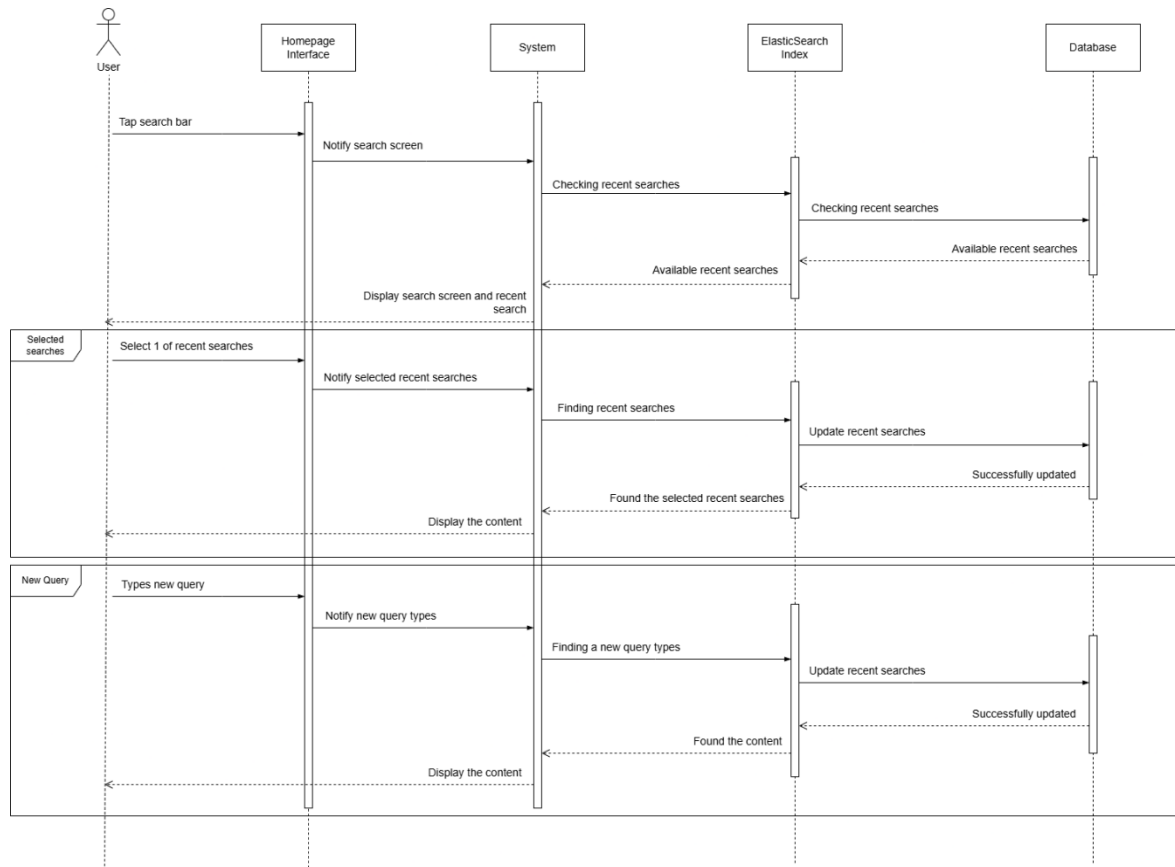


Figure 2. 19 Sequence Search

5. Sequence Message

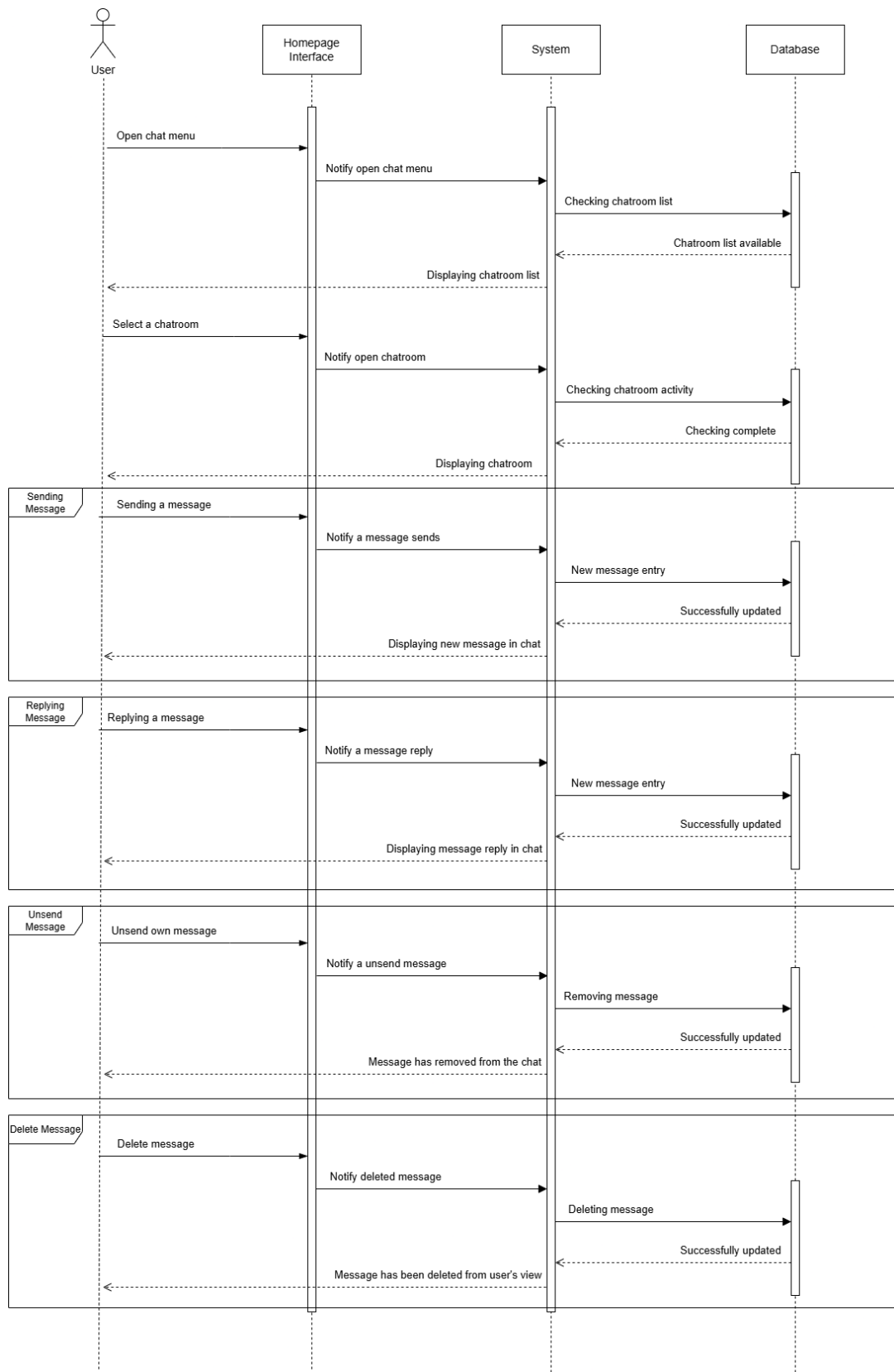


Figure 2. 20 Sequence Message

6. Sequence Self's Profile

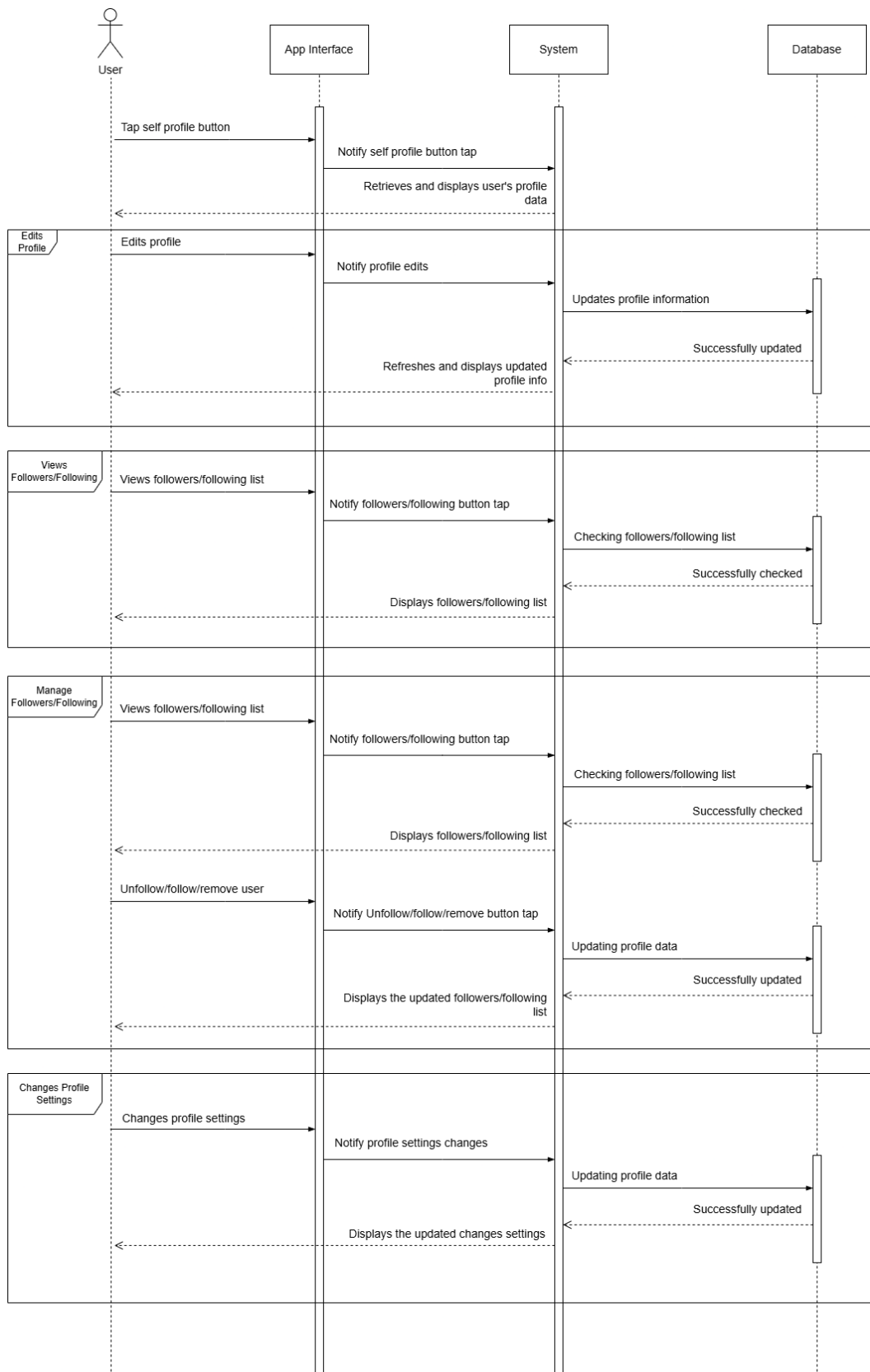


Figure 2. 21 Sequence Self's Profile

7. Sequence Other's Profile

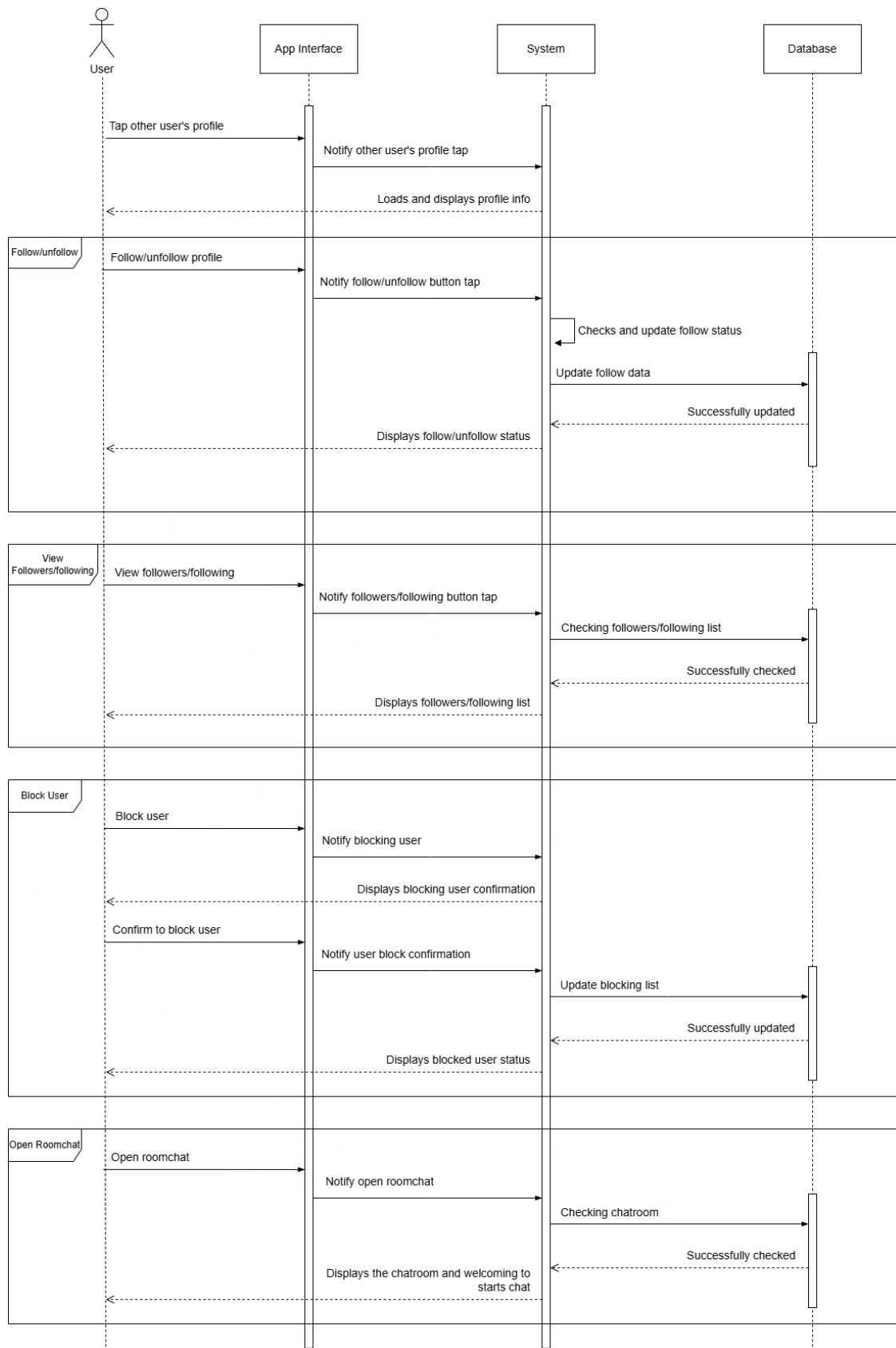


Figure 2. 22 Flowchart Other's Profile

2.6 Class Diagram

The class diagram illustrates the system's object structure and how different classes are related. It shows the attributes and methods of each class, as well as the relationships (e.g., inheritance, associations).

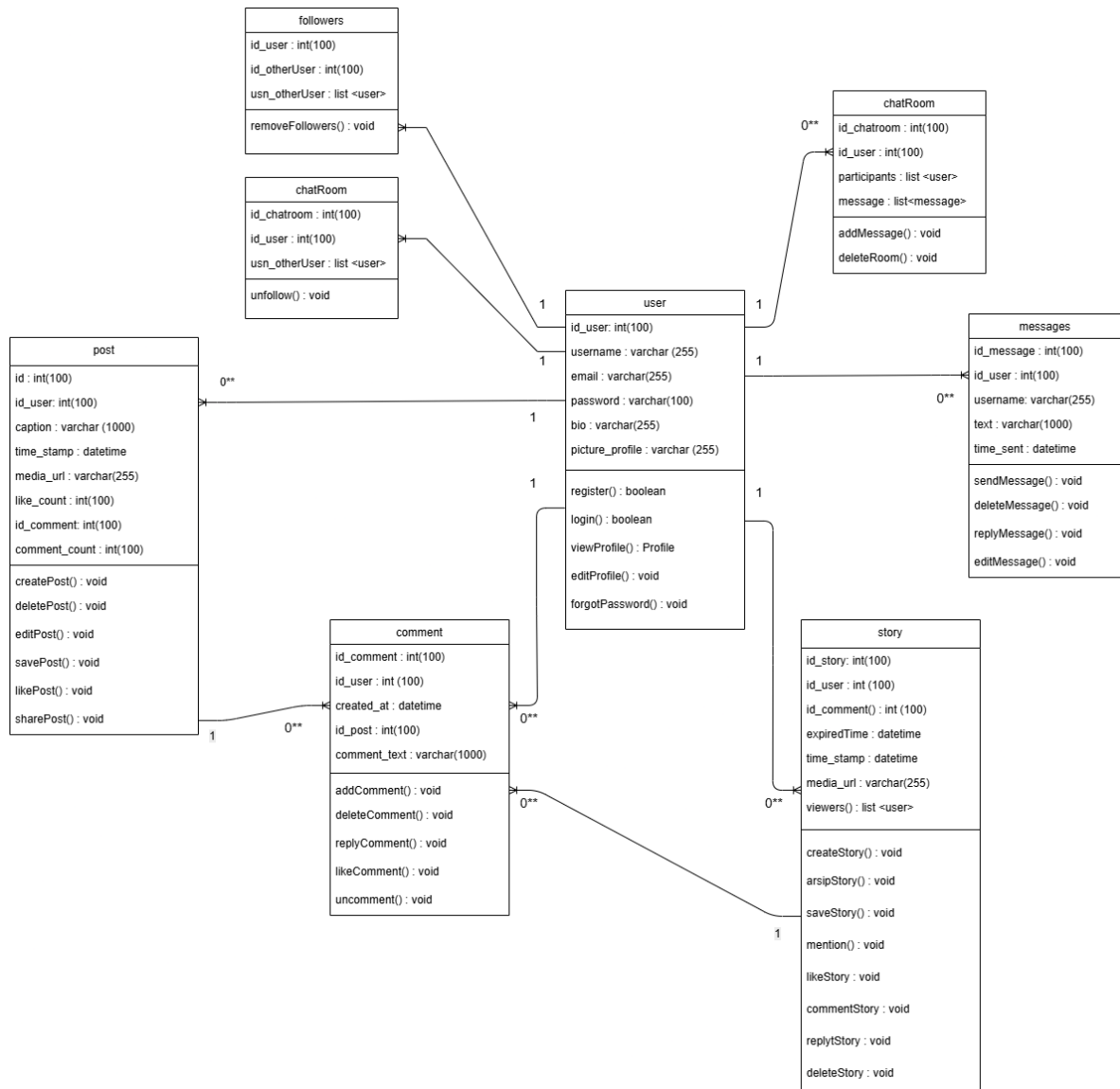


Figure 2. 23 Class Diagram

CHAPTER 3

PROTOTYPE

The Homo Connectus prototype is designed to be intuitive and user-friendly, mimicking the simple yet effective design principles found in popular social media platforms like Instagram.

The key features of the prototype include:

1. **Login Screen:** The first point of interaction for users to access the app. It includes fields for entering the username and password, as well as buttons for login and registration.
2. **Dashboard/Feed Screen:** Displays posts from users that the logged-in user follows. Users can interact with posts through likes, comments, and sharing.
3. **Post Creation Screen:** Allows users to upload images, create text-based posts, or both. It also includes options for adding captions, tags, and hashtags.
4. **Profile Page:** Shows the user's information and content, such as posts and followers. The profile page also allows users

3.1 Design

The prototype for Homo Connectus was developed with an emphasis on simplicity, ease of navigation, and responsiveness across devices. The design follows material design principles, ensuring a clean, user-friendly interface with a focus on accessibility and user engagement.

Main Pages and Features

1. Sign Up Page:

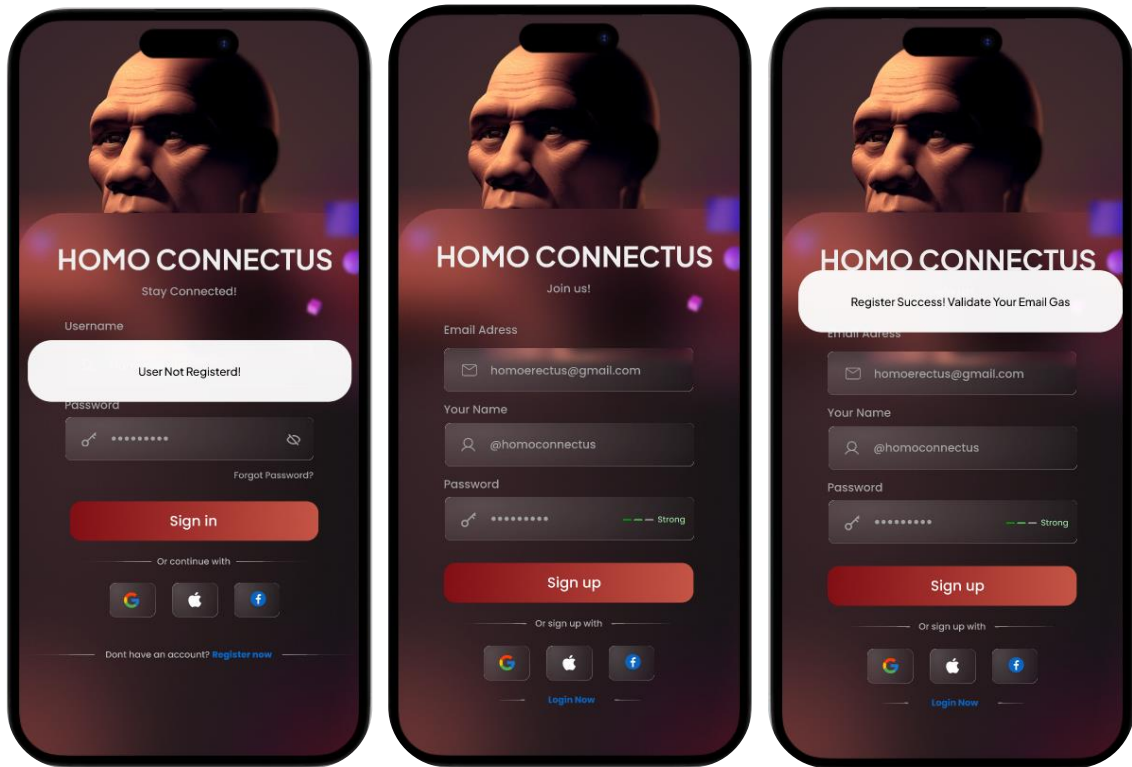


Figure 3. 1 Sign Up Page

- Users who don't have an account can create a new one by providing their username, email, password, and other required details.
- The sign up page includes validation for the input fields (e.g., checking if the email is valid or if the password meets the criteria).
- Upon successful sign-up, the user will be redirected to the Dashboard.

2. Login Page:

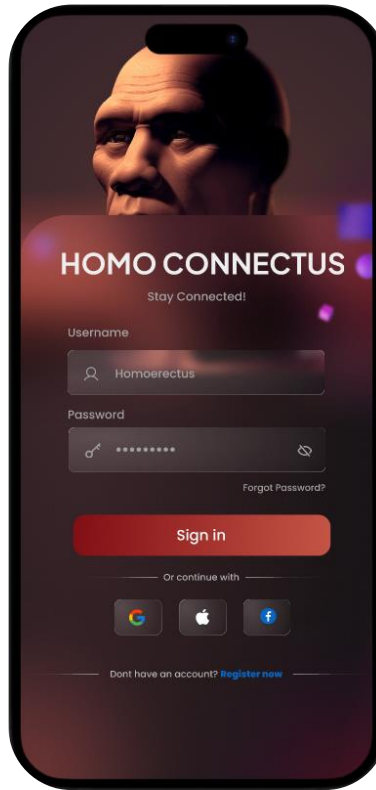


Figure 3. 2 Login Page

- The login page is the first screen the user interacts with. It allows users to input their username and password to gain access to the application.
- The design is minimalist, with a large text field for entering the username and password, and a prominent login button for submission.
- Error handling is implemented to provide feedback if the login credentials are incorrect.

3. Dashboard (Home Feed):



Figure 3. 3 Home Feed

- After logging in, the user is directed to the dashboard, which displays the home feed with posts from users they follow.
- The dashboard allows the user to:
 - Like posts with a heart icon.
 - Comment on posts.
 - Follow/Unfollow users.
- Post creation and user profile options are available through buttons at the top or bottom of the screen.
- The feed layout is vertical, displaying posts with large images or text and a simple button interface.

4. Profile Page:

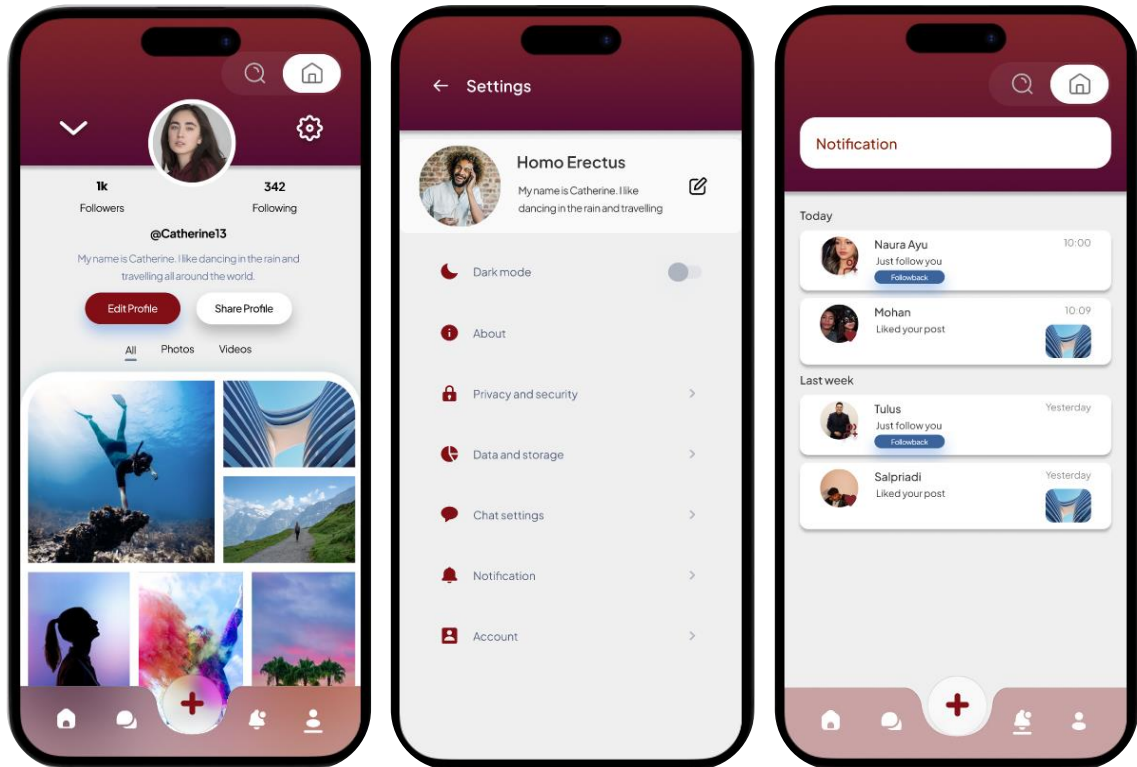


Figure 3. 4 Profile Page

The profile page displays the user's information (name, bio, etc.), their posts, followers, and following.

- The user can edit their profile details and view the posts they have created.
- A settings button allows for account management, including password changes and other preferences.

5. Search Page:

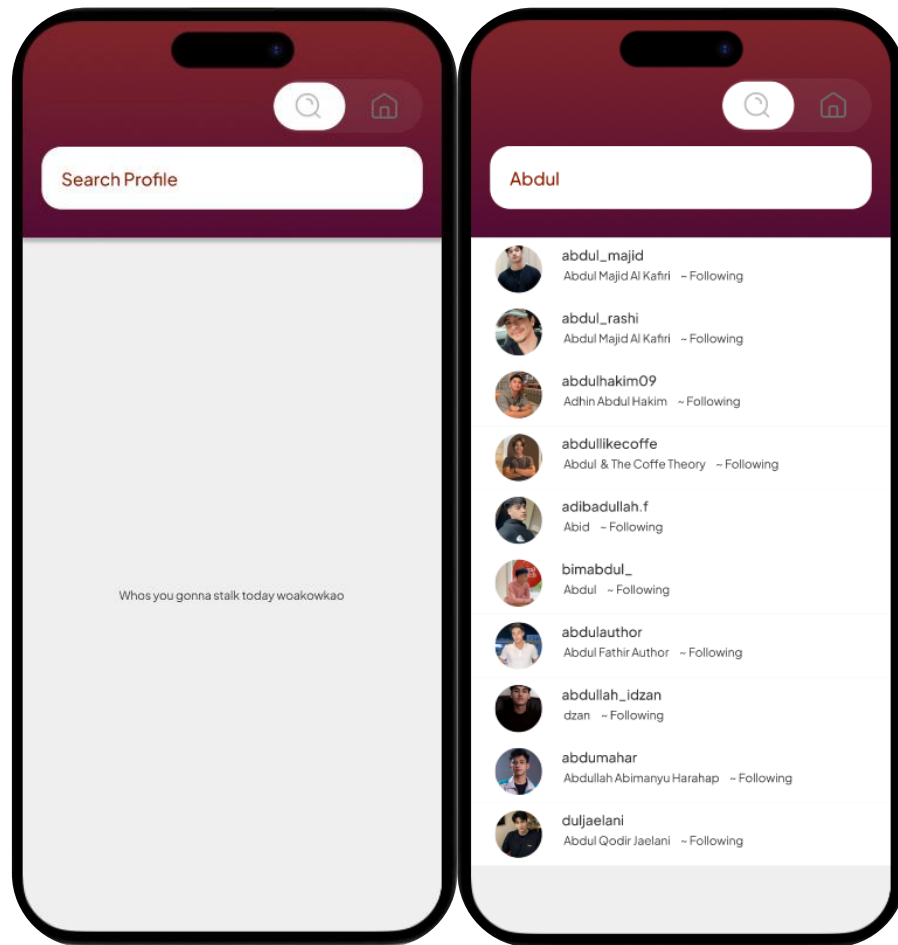


Figure 3. 5 Search Page

- The search page enables users to search for other users or specific posts using keywords or hashtags.
- A search bar is prominently displayed at the top of the screen, and search results are dynamically updated based on user input.

6. Messaging:

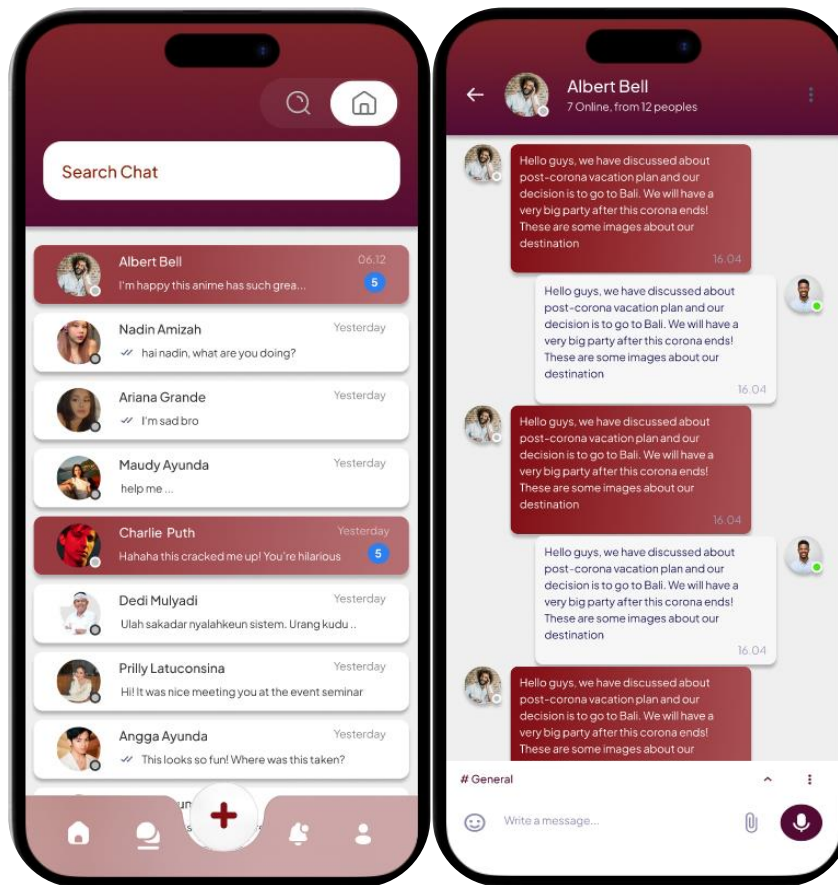


Figure 3. 6 Chat Page

- Users can send direct messages to one another.
- The messaging feature is represented by a message icon on each user's profile page and can be accessed in the chat section.
- The inbox contains a list of conversations, with the latest messages appearing at the top.
- Users can start a new conversation by typing and sending messages in a text input field.
- Conversations are displayed in a chat bubble format, with timestamps and the user's name.

CHAPTER 4

CONCLUSION

4.1 Summary

The Homoconnectus project successfully demonstrated the process of designing and developing a social media-inspired platform using object-oriented programming (OOP) principles. The app, modeled after Instagram, incorporates essential social media functionalities such as:

- User authentication (login and sign-up).
- Post creation, sharing, and interacting with other posts (like, comment, share).
- Profile management and exploring other users' profiles.
- Direct messaging to allow users to communicate privately.

The prototype was created using Figma, providing a user-friendly, responsive design that adapts across devices. By implementing real-time interactions, such as liking and commenting on posts, and enabling direct messaging, Homoconnectus offers an engaging and interactive experience.

4.2 Final Thoughts

The Homoconnectus project was an enriching experience that provided practical insights into designing and building a social media application using modern development techniques. By focusing on essential social media functionalities and user-centered design, we were able to create an engaging and interactive prototype. The project not only enhanced our technical skills but also improved our understanding of how to build user-friendly applications that meet real-world needs.

The journey of creating Homoconnectus has been a stepping stone for future projects, and we are excited about the potential improvements and features that could be added to this platform in the future.